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KULLIYYAH OF INFORMATION & COMMUNICATION TECHNOLOGY

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SECTION 1

ASSIGNMENT 3

TITLE: DESCRIPTIVE ANALYSIS OF DEATH STATISTICS IN MALAYSIA FROM 2001 TO 2021

GROUP 5

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1.0 DATASET SELECTION

This assignment will use the Death by State, Sex and Age Group Malaysia data set, sourced from the Department of Statistics Malaysia (DOSM) via data.gov.my. The data set is suitable for study due to the fact that it covers the official death records recorded in Malaysia between 2001 and 2021 for 21 years, which provides long term data for the study of death patterns and changes over time.

The data set has 13000+ rows and contains the following five key variables which are Year, State, Age Group, Sex and Number Of Death. The variables are adequate for descriptive analysis and visualization. The Year variable can be used to observe the pattern of deaths over the years and the State, Sex and Age Group variables can be used to compare the demographic groups and regions of Malaysia.

Furthermore, the dataset has been deemed reliable and relevant as it was released by DOSM, an official government statistical agency in Malaysia. The volume of data and detailed categorisation aids in identifying meaningful insights, variations and trends in death statistics. Thus, the data set contains all data that is suitable for descriptive analyses and data visualization.

2.0 RESEARCH QUESTION

“What are the demographic and regional trends in deaths statistics in Malaysia from 2001 to 2021?”

3.0 RESEARCH OBJECTIVES

1. To identify the greatest total numbers of deaths that occurred in the year between 2001 and 2021.
2. To find out which is the state with the most death rate in Malaysia.
3. To observe which age group had the maximum death tolls in Malaysia.
4. To investigate the proportionate distribution of recorded deaths by sex.

4.0 DATA PREPROCESSING & VARIABLE SELECTION

4.1 PRE-PROCESSING

The dataset was checked for missing values, duplicates, and inconsistencies before the analysis process. No duplicate records or major outliers were found in the dataset. However, several preprocessing steps were performed to improve data consistency and readability.

1. All column names and character data were converted into uppercase format to standardize the data.

```
> head(data)
> colnames(data) <- toupper(colnames(data))
> data[] <- lapply(data, function(x) { if(is.character(x)) toupper(x) else x })
> head(data)
```

```
> head(data)
  Year    State    Age.group    Sex Number.of.death
1 2001 Malaysia Under 1 year   Male            1817
2 2001 Malaysia Under 1 year Female            1449
3 2001 Malaysia        1-Apr   Male             570
4 2001 Malaysia        1-Apr Female             463
5 2001 Malaysia        5-Sep   Male             501
6 2001 Malaysia        5-Sep Female             324
> colnames(data) <- toupper(colnames(data))
> data[] <- lapply(data, function(x) { if(is.character(x)) toupper(x) else x })
> head(data)
  YEAR    STATE    AGE.GROUP    SEX NUMBER.OF.DEATH
1 2001 MALAYSIA UNDER 1 YEAR   MALE            1817
2 2001 MALAYSIA UNDER 1 YEAR FEMALE            1449
3 2001 MALAYSIA        1-APR   MALE             570
4 2001 MALAYSIA        1-APR FEMALE             463
5 2001 MALAYSIA        5-SEP   MALE             501
6 2001 MALAYSIA        5-SEP FEMALE             324
```

2. Transforming “AGE.GROUP” and “NUMBER.GROUP” into more suitable formats which are “AGE GROUP” and “NUMBER GROUP”.

```
> colnames(data)
> names(data)[names(data) == "AGE.GROUP"] <- "AGE GROUP"
> names(data)[names(data) == "NUMBER.OF.DEATH"] <- "NUMBER OF DEATH"
> colnames(data)
```

```
> colnames(data)
[1] "YEAR"            "STATE"           "AGE GROUP"       "SEX"
[5] "NUMBER OF DEATH"
```

3. True the identification of missing values under the category “NUMBER OF DEATH” from the state of “WP PUTRAJAYA” from 2001 until 2009 turning them into 0 to ensure completeness of the analysis.

```
> data[data$STATE == "WP PUTRAJAYA" & data$YEAR >= 2001 & data$YEAR <= 2010,
c("YEAR", "STATE", "NUMBER OF DEATH")]
> data$`NUMBER OF DEATH`[
+   data$STATE == "WP PUTRAJAYA" &
+   data$YEAR >= 2001 &
+   data$YEAR <= 2009 &
+   is.na(data$`NUMBER OF DEATH`)
+ ] <- 0
> data[data$STATE == "WP PUTRAJAYA" & data$YEAR >= 2001 & data$YEAR <= 2010,
c("YEAR", "STATE", "NUMBER OF DEATH")]
```

```
> data[data$STATE == "WP PUTRAJAYA" & data$YEAR >= 2001 & data$YEAR <= 2010, c("YEAR", "STATE", "NUMBER OF DEATH")]
     YEAR         STATE NUMBER OF DEATH
609  2001 WP PUTRAJAYA              NA
610  2001 WP PUTRAJAYA              NA
611  2001 WP PUTRAJAYA              NA
612  2001 WP PUTRAJAYA              NA
613  2001 WP PUTRAJAYA              NA
614  2001 WP PUTRAJAYA              NA
615  2001 WP PUTRAJAYA              NA
616  2001 WP PUTRAJAYA              NA
617  2001 WP PUTRAJAYA              NA
618  2001 WP PUTRAJAYA              NA
619  2001 WP PUTRAJAYA              NA
620  2001 WP PUTRAJAYA              NA
```

```
> data$`NUMBER OF DEATH`[
+   data$STATE == "WP PUTRAJAYA" &
+   data$YEAR >= 2001 &
+   data$YEAR <= 2009 &
+   is.na(data$`NUMBER OF DEATH`)
+ ] <- 0
```

```
> data[data$STATE == "WP PUTRAJAYA" & data$YEAR >= 2001 & data$YEAR <= 2010, c("YEAR", "STATE", "NUMBER OF DEATH")]
     YEAR         STATE NUMBER OF DEATH
609  2001 WP PUTRAJAYA               0
610  2001 WP PUTRAJAYA               0
611  2001 WP PUTRAJAYA               0
612  2001 WP PUTRAJAYA               0
613  2001 WP PUTRAJAYA               0
614  2001 WP PUTRAJAYA               0
615  2001 WP PUTRAJAYA               0
616  2001 WP PUTRAJAYA               0
617  2001 WP PUTRAJAYA               0
618  2001 WP PUTRAJAYA               0
619  2001 WP PUTRAJAYA               0
620  2001 WP PUTRAJAYA               0
```

4. Correcting the inconsistency in age group labels. The value 1-APR to 1-4, 5-SEP to 5-9 and OCT-14 to 10-14.

```
> unique(data$`AGE GROUP`)
> data$`AGE GROUP`[data$`AGE GROUP` == "1-APR"] <- "1-4"
> data$`AGE GROUP`[data$`AGE GROUP` == "5-SEP"] <- "5-9"
> data$`AGE GROUP`[data$`AGE GROUP` == "OCT-14"] <- "10-14"
> unique(data$`AGE GROUP`)
```

```
> unique(data$`AGE GROUP`)
 [1] "UNDER 1 YEAR" "1-4"          "5-9"          "10-14"        "15-19"
 [6] "20-24"        "25-29"        "30-34"        "35-39"        "40-44"
[11] "45-49"        "50-54"        "55-59"        "60-64"        "65-69"
[16] "70-74"        "75-79"        "80-84"        "85 +"
```

4.2 DATA WRANGLING

Next, data wrangling will be done to the dataset to convert it into a usable format for analysis.

1. First, as the AGE GROUP has too many age brackets (19 brackets), we classify it into 3 separate categories by referring to Department of Statistics Malaysia (DOSM) which is Children from "UNDER 1 YEAR" until "10-14" (0-14), Adults from "15-19" to "60-64" (15-64) and Elderly from "65-69" to "85 +" (65+).

```
> data <- data %>%
+   mutate(
+     `AGE CATEGORY` = case_when(
+       `AGE GROUP` %in% c("UNDER 1 YEAR", "1-4", "5-9", "10-14") ~ "0-14 (CHILDREN)",
+       `AGE GROUP` %in% c("15-19", "20-24", "25-29", "30-34", "35-39", "40-44", "45-49",
"50-54", "55-59", "60-64") ~ "15-64 (ADULTS)",
+       `AGE GROUP` %in% c("65-69", "70-74", "75-79", "80-84", "85 +") ~ "65+ (ELDERLY)",
+       TRUE ~ "UNKNOWN"
+     )
> glimpse(data)
```

```
> glimpse(data)
Rows: 13,566
Columns: 6
$ YEAR             <int> 2001, 2001, 2001, 2001, 2001, 2001, 2001, 2001, 2001…
$ STATE            <chr> "MALAYSIA", "MALAYSIA", "MALAYSIA", "MALAYSIA", "MAL…
$ `AGE GROUP`      <chr> "UNDER 1 YEAR", "UNDER 1 YEAR", "1-4", "1-4", "5-9",…
$ SEX              <chr> "MALE", "FEMALE", "MALE", "FEMALE", "MALE", "FEMALE"…
$ `NUMBER OF DEATH` <dbl> 1817, 1449, 570, 463, 501, 324, 518, 291, 1433, 460,…
$ `AGE CATEGORY`   <chr> "0-14 (CHILDREN)", "0-14 (CHILDREN)", "0-14 (CHILDRE…
```

```
> data %>%
+   select(`AGE GROUP`, `AGE CATEGORY`) %>%
+   distinct() %>%
```

```
+   head(10)
```

```
+   head(10)
      AGE GROUP    AGE CATEGORY
1  UNDER 1 YEAR 0-14 (CHILDREN)
2           1-4 0-14 (CHILDREN)
3           5-9 0-14 (CHILDREN)
4         10-14 0-14 (CHILDREN)
5         15-19  15-64 (ADULTS)
6         20-24  15-64 (ADULTS)
7         25-29  15-64 (ADULTS)
8         30-34  15-64 (ADULTS)
9         35-39  15-64 (ADULTS)
10        40-44  15-64 (ADULTS)
```

2. Next, for STATE, as it also includes Malaysia in STATE's column, we will filter it out into two groups which are states and Malaysia.

```
> df_national <- data %>% filter(STATE == "MALAYSIA")
> df_states <- data %>% filter(STATE != "MALAYSIA")
> unique(df_national$STATE)
> head(unique(df_states$STATE), 5)
```

```
> unique(df_national$STATE)
[1] "MALAYSIA"
> head(unique(df_states$STATE), 5)
[1] "JOHOR"           "KEDAH"           "KELANTAN"        "MELAKA"          "NEGERI SEMBILAN"
```

3. Lastly, we convert the variables into proper data types (integers for calculations and factors for grouping) to ensure accurate descriptive statistics and smooth plotting in R.

```
> data <- data %>%
+   mutate(
+     YEAR = as.integer(YEAR),
+     STATE = as.factor(STATE),
+     `AGE GROUP` = as.factor(`AGE GROUP`),
+     `AGE CATEGORY` = as.factor(`AGE CATEGORY`),
+     SEX = as.factor(SEX),
+     `NUMBER OF DEATH` = as.integer(`NUMBER OF DEATH`)
+   )
> # Check the class/type of each column specifically
> sapply(data, class)
```

```
> sapply(data, class)
          YEAR          STATE      AGE GROUP            SEX NUMBER OF DEATH    AGE CATEGORY
     "integer"       "factor"       "factor"       "factor"       "integer"       "factor"
```

4.3 FEATURE AND VARIABLE SELECTION

Now we will select suitable variable for every Research Objectives

```
> # Dataset for Research Objective 1: Yearly National Totals
> dataset_ro1 <- df_national %>%
+   select(YEAR, `NUMBER OF DEATH`)
> # Dataset for Research Objective 2: Total Death Comparisons by State
> dataset_ro2 <- df_states %>%
+   select(STATE, `NUMBER OF DEATH`)
> # Dataset for Research Objective 3: Death Distributions by Age Category
> dataset_ro3 <- df_national %>%
+   select(`AGE CATEGORY`, `NUMBER OF DEATH`)
> # Dataset for Research Objective 4: Death Distributions by Sex
> dataset_ro4 <- df_national %>%
+   select(SEX, `NUMBER OF DEATH`)
```

```
> # Dataset for Research Objective 1: Yearly National Totals
> dataset_ro1 <- df_national %>%
+   select(YEAR, `NUMBER OF DEATH`)
>
> # Dataset for Research Objective 2: Total Death Comparisons by State
> dataset_ro2 <- df_states %>%
+   select(STATE, `NUMBER OF DEATH`)
>
> # Dataset for Research Objective 3: Death Distributions by Age Category
> dataset_ro3 <- df_national %>%
+   select(`AGE CATEGORY`, `NUMBER OF DEATH`)
>
> # Dataset for Research Objective 4: Death Distributions by Sex
> dataset_ro4 <- df_national %>%
+   select(SEX, `NUMBER OF DEATH`)
```

Finally we will put them into table

a. YEAR

```
> # Calculate total deaths per year
> total_deaths_year <- dataset_ro1 %>%
+   group_by(YEAR) %>%
+   summarise(TOTAL_DEATHS = sum(`NUMBER OF DEATH`, na.rm = TRUE))
> # View results
> print(total_deaths_year, n = 21)
```

```
> # Calculate total deaths per year
> total_deaths_year <- dataset_ro1 %>%
+   group_by(YEAR) %>%
+   summarise(TOTAL_DEATHS = sum(`NUMBER OF DEATH`, na.rm = TRUE))
>
> # View results
> print(total_deaths_year, n = 21)
# A tibble: 21 × 2
    YEAR TOTAL_DEATHS
   <int>        <dbl>
 1  2001       104531
 2  2002       110367
 3  2003       112744
 4  2004       113192
 5  2005       115436
 6  2006       117778
 7  2007       120670
 8  2008       127098
 9  2009       131328
10  2010       130978
11  2011       135463
12  2012       138692
13  2013       142202
14  2014       150318
15  2015       155786
16  2016       162201
17  2017       168168
18  2018       172031
19  2019       173746
20  2020       166507
21  2021       224569
```

b. STATE

```
> # Calculate total deaths per state (sorted highest to lowest)
> total_deaths_state <- dataset_ro2 %>%
+   group_by(STATE) %>%
+   summarise(TOTAL_DEATHS = sum(`NUMBER OF DEATH`, na.rm = TRUE)) %>%
+   arrange(desc(TOTAL_DEATHS))
> # View results
> print(total_deaths_state)
```

```
> # Calculate total deaths per state (sorted highest to lowest)
> total_deaths_state <- dataset_ro2 %>%
+   group_by(STATE) %>%
+   summarise(TOTAL_DEATHS = sum(`NUMBER OF DEATH`, na.rm = TRUE)) %>%
+   arrange(desc(TOTAL_DEATHS))
>
> # View results
> print(total_deaths_state)
# A tibble: 16 × 2
   STATE           TOTAL_DEATHS
   <chr>                  <dbl>
 1 SELANGOR              441068
 2 JOHOR                 364700
 3 PERAK                 342437
 4 KEDAH                 258129
 5 SARAWAK               240844
 6 KELANTAN              210749
 7 SABAH                 204116
 8 PULAU PINANG          196688
 9 PAHANG                162875
10 WP KUALA LUMPUR       151581
11 TERENGGANU            129967
12 NEGERI SEMBILAN       128368
13 MELAKA                 99360
14 PERLIS                 35420
15 WP LABUAN               5308
16 WP PUTRAJAYA            2195
```

c. AGE

```
> # Calculate total deaths by broad age category
> total_deaths_age <- dataset_ro3 %>%
+   group_by(`AGE CATEGORY`) %>%
+   summarise(TOTAL_DEATHS = sum(`NUMBER OF DEATH`, na.rm = TRUE))
> # View results
> print(total_deaths_age)
```

```
> # Calculate total deaths by broad age category
> total_deaths_age <- dataset_ro3 %>%
+   group_by(`AGE CATEGORY`) %>%
+   summarise(TOTAL_DEATHS = sum(`NUMBER OF DEATH`, na.rm = TRUE))
>
> # View results
> print(total_deaths_age)
# A tibble: 3 × 2
  `AGE CATEGORY`  TOTAL_DEATHS
  <chr>                  <dbl>
1 0-14 (CHILDREN)       115018
2 15-64 (ADULTS)       1226025
3 65+ (ELDERLY)        1632762
```

d. SEX

```
> # Calculate total deaths by sex
> total_deaths_sex <- dataset_ro4 %>%
+   group_by(SEX) %>%
+   summarise(TOTAL_DEATHS = sum(`NUMBER OF DEATH`, na.rm = TRUE))
> # View results
> print(total_deaths_sex)
```

```
> # Calculate total deaths by sex
> total_deaths_sex <- dataset_ro4 %>%
+   group_by(SEX) %>%
+   summarise(TOTAL_DEATHS = sum(`NUMBER OF DEATH`, na.rm = TRUE))
>
> # View results
> print(total_deaths_sex)
# A tibble: 2 × 2
  SEX    TOTAL_DEATHS
  <chr>         <dbl>
1 FEMALE      1262175
2 MALE        1711630
```

5.0 DESCRIPTIVE ANALYSTICS & DATA VISUALIZATION

a. YEAR

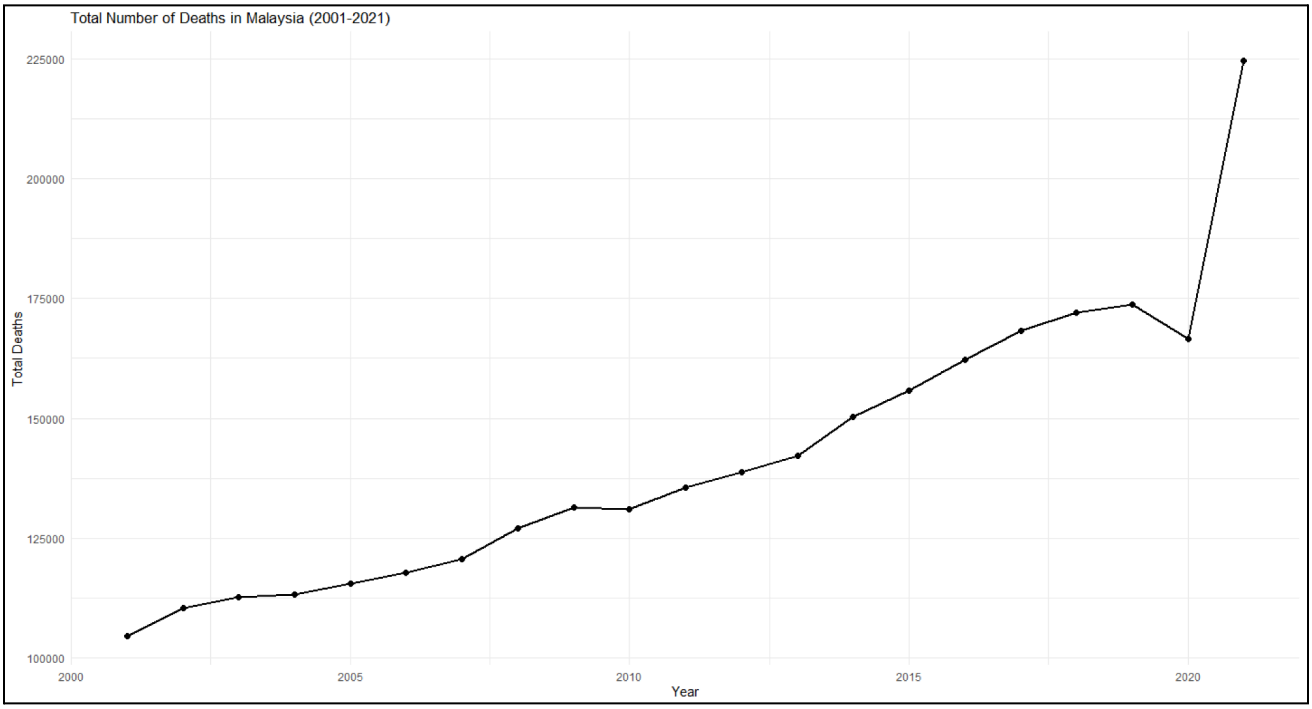
```
# 1. Statistical Measures for Year
> mean(total_deaths_year$TOTAL_DEATHS)
> median(total_deaths_year$TOTAL_DEATHS)
> sd(total_deaths_year$TOTAL_DEATHS)
> table_year <- total_deaths_year %>%
+   mutate(
+     PERCENTAGE = round((TOTAL_DEATHS / sum(TOTAL_DEATHS)) * 100, 2)
+   )
> print("--- Year Descriptive Table ---")
> print(table_year, n = Inf)
```

```
> mean(total_deaths_year$TOTAL_DEATHS)
[1] 141609.8
> median(total_deaths_year$TOTAL_DEATHS)
[1] 135463
> sd(total_deaths_year$TOTAL_DEATHS)
[1] 29176.76
> table_year <- total_deaths_year %>%
+    mutate(
+      PERCENTAGE = round((TOTAL_DEATHS / sum(TOTAL_DEATHS)) * 100, 2)
+   )
> print("--- Year Descriptive Table ---")
[1] "--- Year Descriptive Table ---"
> print(table_year, n = Inf)
# A tibble: 21 × 3
    YEAR TOTAL_DEATHS PERCENTAGE
   <int>        <dbl>      <dbl>
 1  2001       104531       3.52
 2  2002       110367       3.71
 3  2003       112744       3.79
 4  2004       113192       3.81
 5  2005       115436       3.88
 6  2006       117778       3.96
 7  2007       120670       4.06
 8  2008       127098       4.27
 9  2009       131328       4.42
10  2010       130978       4.4
11  2011       135463       4.56
12  2012       138692       4.66
13  2013       142202       4.78
14  2014       150318       5.05
15  2015       155786       5.24
16  2016       162201       5.45
17  2017       168168       5.65
18  2018       172031       5.78
19  2019       173746       5.84
20  2020       166507       5.6
21  2021       224569       7.55
```

The total number of deaths in Malaysia increased from 2001 to 2021. The analysis showed the number of recorded deaths in 2001 was 104,531 while the highest total number of deaths was recorded in 2021 with 224.569 deaths, which also represents 7.55% of the overall dataset. Between 2001 and 2021, there were 141,609.8 deaths on average, with a median of 135,463. The yearly death totals vary noticeably over time, as indicated by the standard deviation of 29,176.76.By determining the year with the highest overall number of deaths and examining the annual pattern of deaths in Malaysia between 2001 and 2021, the greatest total number of deaths is able to be recognized.

```
# Line chart for Total Deaths by Year
> ggplot(total_deaths_year,
+        aes(x = YEAR, y = TOTAL_DEATHS)) +
+   geom_line(size = 1) +
+   geom_point(size = 2) +
+   labs(
+     title = "Total Number of Deaths in Malaysia (2001-2021)",
+     x = "Year",
+     y = "Total Deaths"
```

```
+   ) +
+   theme_minimal()
```



The line chart visualizes the pattern of total deaths in Malaysia from 2001 to 2021. The chart clearly shows that the total deaths are inclining over the years.There was a slight decrease in 2020 before it spiked in 2021. The highest total of deaths was recorded in 2021 compared to the years before it. From the analyzed data, an upward trend was discovered in the total deaths, especially after 2014. The high increase in 2021 may be factored by population growth, aging population, and the public's health during the time. Therefore, the visualisation successfully supports Research Objective 1 by clearly identifying the year with the greatest total number of deaths between 2001 and 2021.

b. STATE

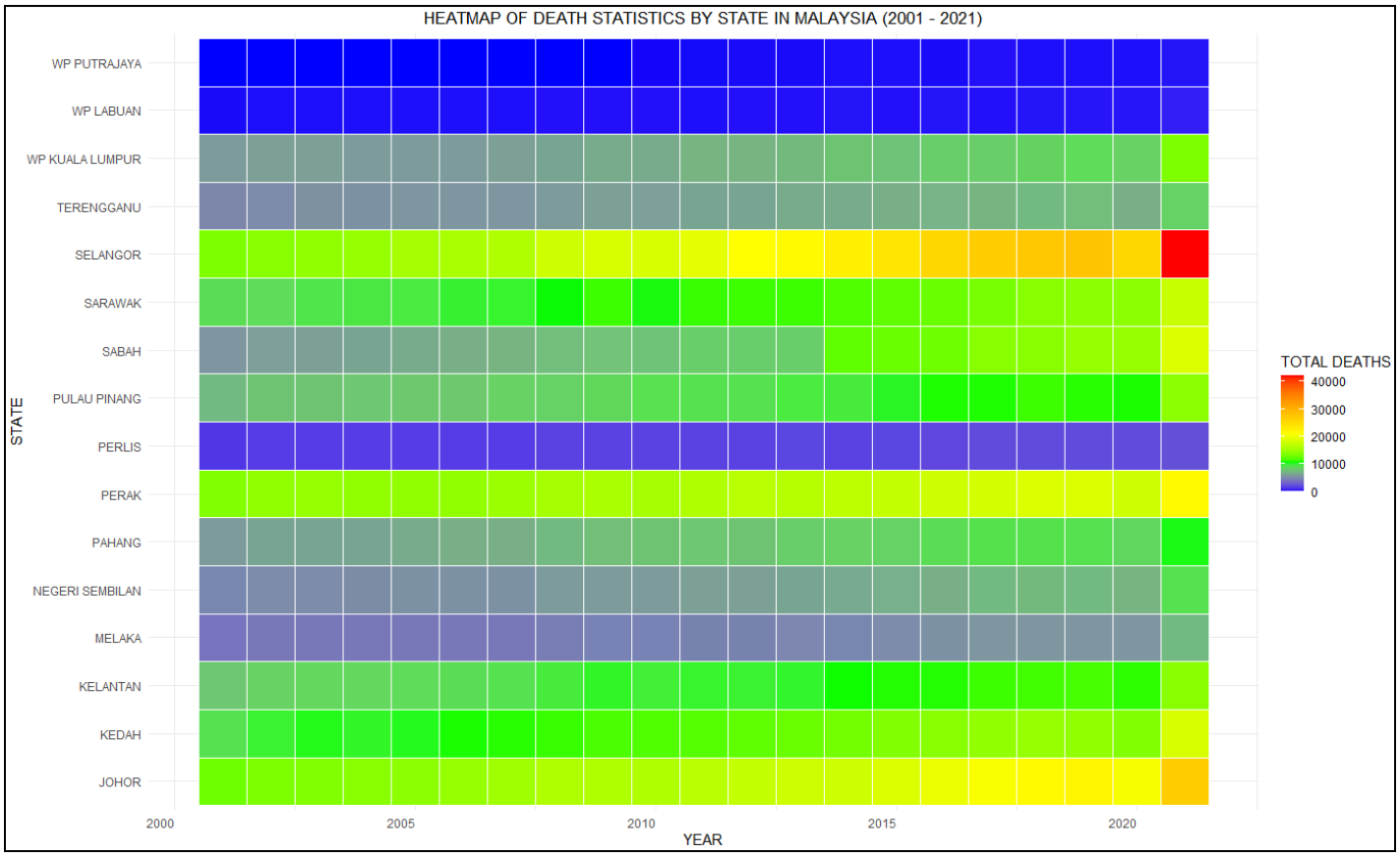
```
# 2. Statistical Measures for State
> mean(total_deaths_state$TOTAL_DEATHS)
> median(total_deaths_state$TOTAL_DEATHS)
> sd(total_deaths_state$TOTAL_DEATHS)
> table_state <- total_deaths_state %>%
+   mutate(PERCENTAGE = round((TOTAL_DEATHS / sum(TOTAL_DEATHS)) * 100, 2))
> print("--- State Descriptive Table ---")
> print(table_state)
```

```
> mean(total_deaths_state$TOTAL_DEATHS)
[1] 185862.8
> median(total_deaths_state$TOTAL_DEATHS)
[1] 179781.5
> sd(total_deaths_state$TOTAL_DEATHS)
[1] 125306.6
> table_state <- total_deaths_state %>%
+   mutate(PERCENTAGE = round((TOTAL_DEATHS / sum(TOTAL_DEATHS)) * 100, 2))
> print("--- State Descriptive Table ---")
[1] "--- State Descriptive Table ---"
> print(table_state)
# A tibble: 16 × 3
   STATE           TOTAL_DEATHS PERCENTAGE
   <chr>                  <dbl>      <dbl>
 1 SELANGOR              441068      14.8
 2 JOHOR                 364700      12.3
 3 PERAK                 342437      11.5
 4 KEDAH                 258129       8.68
 5 SARAWAK               240844       8.1
 6 KELANTAN              210749       7.09
 7 SABAH                 204116       6.86
 8 PULAU PINANG          196688       6.61
 9 PAHANG                162875       5.48
10 WP KUALA LUMPUR       151581       5.1
11 TERENGGANU            129967       4.37
12 NEGERI SEMBILAN       128368       4.32
13 MELAKA                 99360       3.34
14 PERLIS                 35420       1.19
15 WP LABUAN               5308       0.18
16 WP PUTRAJAYA            2195       0.07
```

The highest total of death cases was recorded in Selangor with 441,068 cases which is 14.8% of the total death cases in the dataset. This suggests that Selangor had the highest percentage of death cases in Malaysia over the period of study. By contrast, WP Putrajaya had the smallest total death cases, that is 2,195 cases or only 0.07% of the total death cases. Furthermore, the mean was 185862.8 and the median of deaths for the states was 179781.5. This implies that in some states the number of deaths exceeded the average value, especially Selangor, Johor and Perak. The standard deviation of 125306.6 also indicates high variability among states in terms of death numbers implying that death statistics were not evenly distributed in Malaysia. The overall result of this analysis can confirm the research objective 2 that shows that Selangor is the state with the most death tolls recorded in Malaysia from the year 2001 to 2021.

```
# Heatmap for Total Deaths by State
> heatmap_state <- ggplot(heatmap_data,
+ aes(x = YEAR,
+ y = STATE,
+ fill = TOTAL_DEATHS)) +
+ geom_tile(color = "white") +
+ scale_fill_gradientn(colors = c("blue", "purple",
+ "green", "yellow", "orange", "red")) +
+ labs(title = "HEATMAP OF DEATH STATISTICS BY STATE IN MALAYSIA (2001 - 2021)",
+ x = "YEAR", y = "STATE", fill = "TOTAL DEATHS") +
+ theme_minimal() +
```

```
+ theme(axis.text.x = element_text(angle = 0, hjust = 1),
+ plot.title = element_text(hjust = 0.5))
> print(heatmap_state)
```



Based on the heatmap, Selangor consistently recorded the highest number of deaths from 2001 to 2021 as indicated by the dominant yellow, orange, and red color intensities compared to other states However, the dark blue colors indicate the lowest number of deaths consistently recorded by the two states of WP Putrajaya and WP Labuan in all years. Other states like Johor, Perak and others had relatively high death rates over time while other smaller states like Perlis and Melaka kept their death rates low throughout the period. The heatmap also shows that death statistics generally increased over time across most states, particularly towards the end of the period analyzed. Several factors can affect this trend, including population growth, urbanization, the aging population, and public health conditions. So, the heatmap succeeded to support the research objective 2 which was to identify Selangor as the state having the largest percentage of recorded deaths from 2001 to 2021.

c. AGE

```
# 3. Statistical Measures for Age Category
mean(total_deaths_age$TOTAL_DEATHS)
median(total_deaths_age$TOTAL_DEATHS)
sd(total_deaths_age$TOTAL_DEATHS)
```

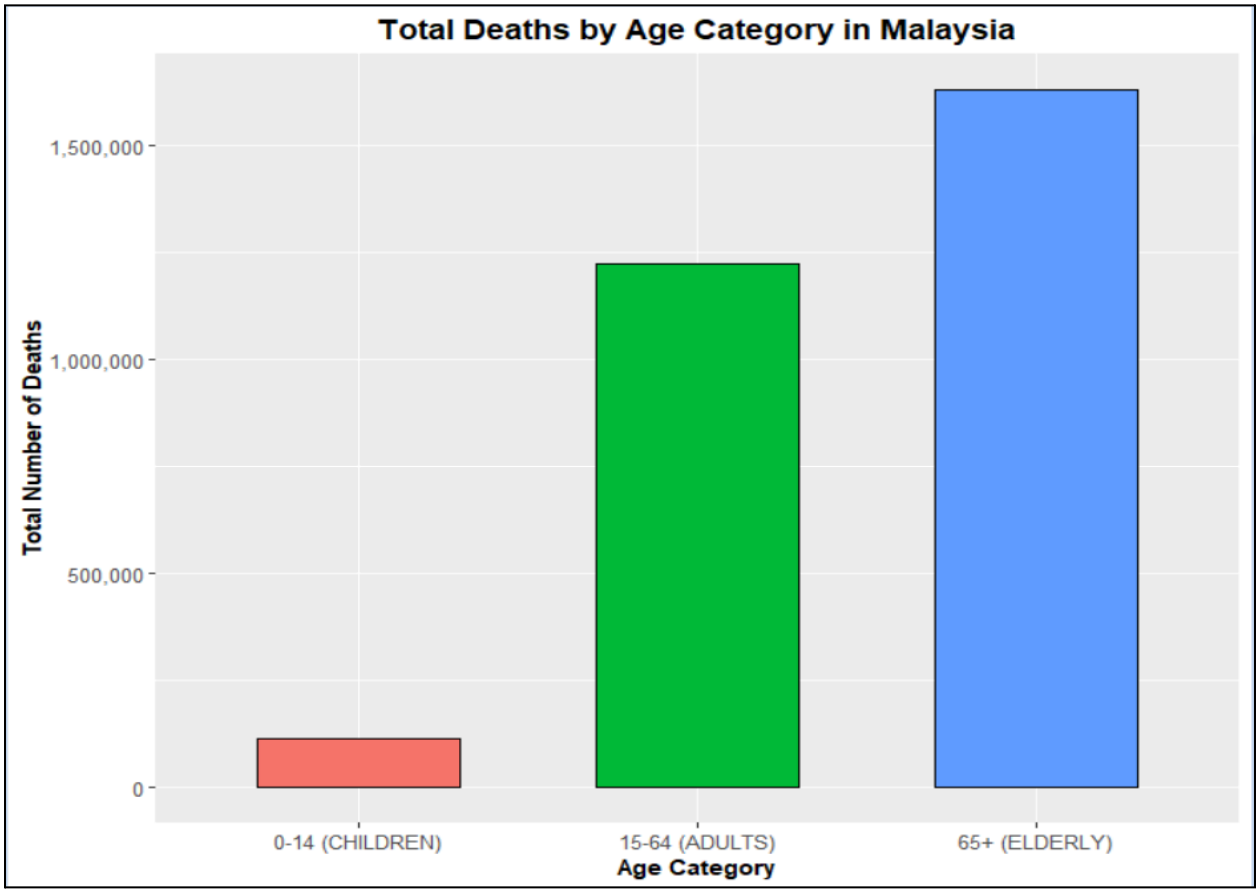
```
table_age <- total_deaths_age %>%
  mutate(PERCENTAGE = round((TOTAL_DEATHS / sum(TOTAL_DEATHS)) * 100, 2))
print("--- Age Category Descriptive Table ---")
print(table_age)
```

```
> mean(total_deaths_age$TOTAL_DEATHS)
[1] 991268.3
> median(total_deaths_age$TOTAL_DEATHS)
[1] 1226025
> sd(total_deaths_age$TOTAL_DEATHS)
[1] 785633.3
> table_age <- total_deaths_age %>%
+   mutate(PERCENTAGE = round((TOTAL_DEATHS / sum(TOTAL_DEATHS)) * 100, 2))
> print("--- Age Category Descriptive Table ---")
[1] "--- Age Category Descriptive Table ---"
> print(table_age)
# A tibble: 3 × 3
  `AGE CATEGORY`  TOTAL_DEATHS PERCENTAGE
  <chr>                  <dbl>      <dbl>
1 0-14 (CHILDREN)       115018       3.87
2 15-64 (ADULTS)       1226025      41.2
3 65+ (ELDERLY)        1632762      54.9
```

The highest percentage of deaths in Malaysia is elderly people where they are 65 years old and above with 54.60%, covering more than half of the total deaths in Malaysia from 2001 to 2021, followed by adults (41.23%) and the lowest is children with 3.87% only. The percentage itself shows us that deaths in Malaysia commonly happen in the older ages. Next, the mean for age category is 991268 while the median is 1226025. As the median is higher than mean, it also proves that the data is shifted towards older ages. We can say that deaths in Malaysia are not spread out evenly, but they are heavily concentrated in the older age groups. As the standard deviation is also very high (785633.3), it shows that there is a big gap in younger and older people's deaths, proving that age is also a major factor in mortality rates.

```
# Bar chart for Total Deaths by Age Category
ggplot(total_deaths_age, aes(x = `AGE CATEGORY`, y = TOTAL_DEATHS, fill = `AGE
CATEGORY`)) +
  geom_bar(stat = "identity", color = "black", width = 0.6) +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Total Deaths by Age Category in Malaysia",
    x = "Age Category",
    y = "Total Number of Deaths",
    fill = "Age Group"
  ) +
  theme(
    plot.title = element_text(face = "bold", size = 14, hjust = 0.5),
    axis.title = element_text(face = "bold", size = 11),
```

```
  axis.text = element_text(size = 10),
  legend.position = "none"
)
```



Based on the bar chart, it is evident that the elderly population (65+) recorded the highest number of deaths in Malaysia from 2001 to 2021, whereas total deaths among children remained exceptionally low. This trend suggests that a significant portion of the population passes away due to age-related factors, while also reflecting the success of healthcare interventions in keeping child mortality low. Consequently, this visualization directly supports Research Objective 3 by demonstrating that mortality in Malaysia is heavily concentrated among the elderly.

d. SEX

```
# 4. Statistical Measures for Sex
nnual_stats <- data %>%
+   filter(STATE %in% c("MALAYSIA", "Malaysia")) %>%
+   mutate(SEX = toupper(SEX)) %>%
+   group_by(YEAR, SEX) %>%
+   summarise(ANNUAL_TOTAL = sum(`NUMBER OF DEATH`, na.rm = TRUE), .groups =
"drop") %>%
+   group_by(SEX) %>%
```

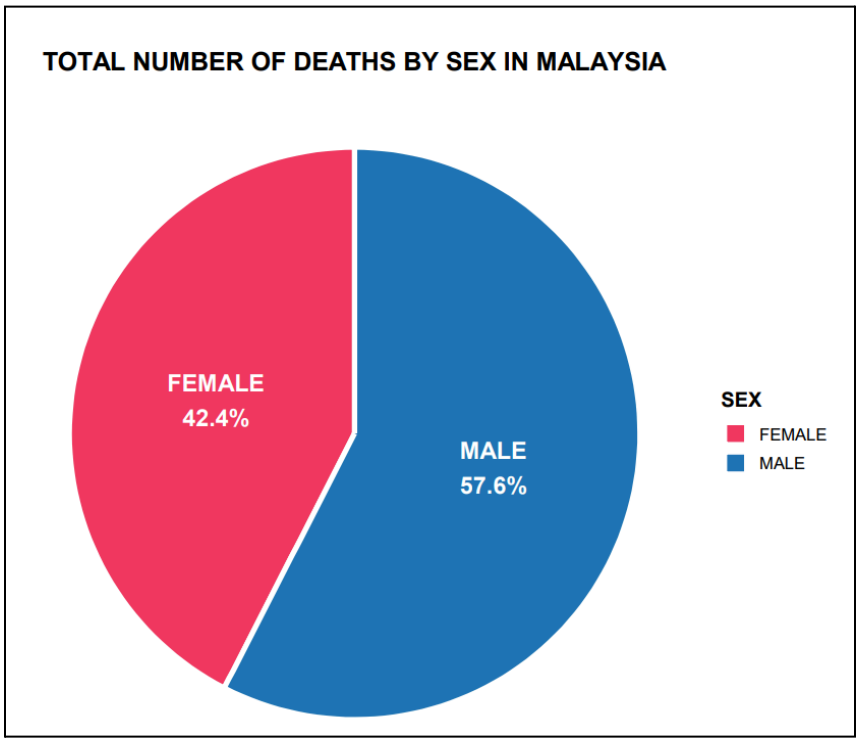
```
+   summarise(
+     MEAN_DEATHS   = mean(ANNUAL_TOTAL),
+     MEDIAN_DEATHS = median(ANNUAL_TOTAL),
+     SD_DEATHS     = sd(ANNUAL_TOTAL),
+     .groups = "drop"
+   )
```

```
> print(gender_annual_stats)
# A tibble: 2 × 4
  SEX    MEAN_DEATHS MEDIAN_DEATHS SD_DEATHS
  <chr>        <dbl>         <int>     <dbl>
1 FEMALE      60104.         57294    12619.
2 MALE        81506.         78169    16566.
```

Annual mortality data reveals a significant and systematic gender difference in Malaysia for the 21 years considered in this study. The categorical mode of the dataset is Male, which also has a significantly higher annual average with a mean of 81,506.2 deaths per year, than the annual average of the Female annual mean of 60,103.6 deaths. The annual medians (78,169.0 for males and 57,294.0 for females) further support this structural imbalance: high mortality rates for males are a regular historical trend, and not a temporary change in the baseline. Additionally, males had a greater range of data dispersion with a standard deviation of 16,565.6 deaths compared to females with a standard deviation of 12,619.1 deaths. This larger absolute change suggests that the male cohort had a much larger year-on-year change, making it more vulnerable to macroeconomic fluctuations, lifestyle risks and the significant epidemiological shocks seen over the course of tracking.

```
# Pie chart for Total Deaths by Sex
> colnames(data) <- toupper(colnames(data))
> colnames(data) <- gsub("\\.", " ", colnames(data))
> pie_data <- data %>%
+   filter(STATE %in% c("MALAYSIA", "Malaysia")) %>%
+   mutate(SEX = toupper(SEX)) %>%
+   group_by(SEX) %>%
+   summarise(TOTAL_DEATHS = sum(`NUMBER OF DEATH`, na.rm = TRUE)) %>%
+   mutate(PERCENTAGE = (TOTAL_DEATHS / sum(TOTAL_DEATHS)) * 100)
> ggplot(pie_data, aes(x = "", y = TOTAL_DEATHS, fill = SEX)) +
+   geom_bar(stat = "identity", width = 1, color = "white", linewidth = 1.5) +
+   coord_polar(theta = "y", start = 0) +
+   scale_fill_manual(values = c("FEMALE" = "#F03760", "MALE" = "#1F77B4")) +
+   labs(title = "TOTAL NUMBER OF DEATHS BY SEX IN MALAYSIA", fill = "SEX") +
+   theme_void() +
+   theme(
+     plot.title = element_text(face = "bold", size = 15, hjust = 0.5),
+     legend.title = element_text(face = "bold", size = 12),
```

```
+    legend.text = element_text(size = 10)
+  ) +
+  geom_text(aes(label = paste0(SEX, "\n", round(PERCENTAGE, 1), "%")),
+            position = position_stack(vjust = 0.5),
+            color = "white", fontface = "bold", size = 5)
```



Based on the pie chart, a significant and enduring gender difference in percentages is evident when the aggregated 21 year period is categorized. Focusing on the categorical mode, the majority of deaths recorded is male accounting for 1711630 (57.6%) of the total deaths recorded in Malaysia during the period of the study with the total recorded deaths of 2973805 during the study. The Female cohort, on the other hand, accounts for the rest of the national distribution at 42.4% (1,262,175 records). The structural breakdown accounts for the clear baseline gender gap of 15.2 % that shows that cumulative male frequencies far exceed female frequencies within the observed time period. There is robust demographic evidence of the long-term vulnerability, behavioral pattern and occupational exposure profile differences between the biological sexes in this enduring baseline imbalance.